

₁ Mechanistic Explainability in Reinforcement
₂ Learning

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The Golem Mythos

A creation that surpasses its creator.

This idea has long captured the imagination and continues to resonate in contemporary society. Long before modern technology, humanity had already questioned whether its creations might escape control.

One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature made of clay and created by humans to serve them [Nocks, 1998]. Figure ?? illustrates the creation of such a servant. The earliest accounts describe the Golem as a protective and helpful figure, reflecting the idea that humans can create an artificial being for the benefit of society.

This vision becomes more ambiguous in the seventeenth century, as illustrated by the legend of the Golem of Prague. In this story, the initially small Golem gradually grows in size and strength until it can no longer be controlled [Nocks, 1998]. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

This narrative motif has been revisited in different forms to illustrate the risks inherent in creation. This continuity is also reflected in the emergence of the modern notion of the robot. The concept of the robot as an artificial servant was introduced in the play *R.U.R. (Rossum's Universal Robots)*, written in 1920 [Čapek, 1920]. In the play, a scientist creates artificial beings designed to perform labour. Conceived as servants, these artificial beings turn against humanity and destroy civilisation. The connection between the Golem and the robot was explicitly made by the author, as illustrated by his statement that “the Robot is the Golem made flesh through mass production” [Comrada, 1995]. Despite their destructive role in the play, the word robot is now used to designate artificial beings created to serve humans. From its origin, the word robot embodies the hopes and fears surrounding technological progress.

Today, this tension between the hopes and fears surrounding technological progress is particularly evident in the development of artificial intelligence. It is therefore not surprising that this field offers an almost literal modern adaptation of the Golem myth. Artificial systems are created to automate tasks traditionally associated with human intelligence. Their rapid development in recent years has transformed a wide range of domains, further intensifying the excitement and concerns surrounding artificial intelligence.

Among the subfields of artificial intelligence, reinforcement learning most directly aims to create autonomous agents. In this learning paradigm, an artificial agent interacts with an environment, selects actions, and receives



Figure 1: The creation of the Golem, by Mikoláš Aleš [Aleš and Wenig, 1908].

44 feedback according to the outcomes of its behaviour [Sutton and Barto, 2018].
 45 Through repeated interaction with its environment, the agent learns a strategy
 46 to achieve the objective. A strength of this approach is its ability to produce
 47 autonomous systems with minimal human intervention. Their agent behavior is
 48 not programmed but learned through interactions with the environment [Amodei
 49 et al., 2016]. This minimal human guidance enables reinforcement learning
 50 systems to achieve superhuman capabilities [Silver et al., 2017, OpenAI et al.,
 51 2019].

52 However, maintaining control over such agents presents a major challenge,
 53 as the very autonomy that makes this learning approach powerful also makes
 54 their behavior more difficult to control. Despite the difficulty of this task,
 55 understanding the behaviour of such agents is therefore the main objective of
 56 this manuscript. This work therefore places itself within a long-standing line of
 57 thought that can be summarized by the following question: how can we ensure
 58 that our creations remain aligned with human intentions?

Chapter 1

Introduction

This initial chapter provides a general introduction to this manuscript. It begins with the research context that motivates this work, followed by the research questions this work addresses. It further presents the resulting contributions and concludes with an overview of the manuscript's structure.

1.1 Research Context

As no one can have failed to notice in recent years, Artificial Intelligence (AI) has advanced rapidly. This progress is driven largely by major developments in Deep Learning (DL). This type of approach relies on artificial Neural Networks (NNs) to learn directly from data [Goodfellow et al., 2016]. These networks are connectionist models in which information is represented and transformed through patterns of interactions among interconnected elementary computational units [Rumelhart et al., 1986]. This property makes neural networks highly versatile, enabling their successful application across an increasing number of domains.

Among the fields that have been transformed by DL is Reinforcement Learning (RL). The purpose of RL is to enable an agent to learn how to make decisions through interactions with an environment [Sutton and Barto, 2018]. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a behavior that maximizes the cumulative rewards obtained over time. Due to their flexibility, NNs provide a powerful means of representing such policies enable agents to learn complex behaviors. Their use has consequently made it possible to apply RL to problems that were previously intractable [Mnih et al., 2013, Silver et al., 2017, OpenAI et al., 2019].

When an agent learns to act in complex environments, it develops strategies that are themselves complex. Although such agents can achieve strong performance, these strategies can be difficult for to understand. From an external perspective, we observe sets of sequences of interactions between the agent and its environment. However, when interactions unfold over long sequences or take place in environments with complex dynamics, these observations are not sufficient to infer the strategy developed by the agent.

An alternative approach is to analyze the agent itself. However, due to

their connectionist nature, NN that model the agent’s policy are not directly interpretable. The information behind a decision is distributed across many units and connections making it impossible to trace. NNs are therefore often regarded as “black box” models [Guidotti et al., 2018].

To address this black-box problem, the field of eXplainable Artificial Intelligence (XAI) has emerged [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In the specific context of RL, this research area is referred to as eXplainable Reinforcement Learning (XRL) [Puiutta and Veith, 2020, Milani et al., 2024]. XRL develops methods for producing explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust and facilitating the adoption of RL agents in real-world settings, identifying errors and biases, and verifying that previously identified problems have been effectively corrected [Barredo Arrieta et al., 2020, Milani et al., 2024].

This manuscript presents XRL methods from different approaches to provide a broader view of the field. Particular emphasis is placed on the mechanistic approach to explainability, which we refer to as Mechanistic Explainability (ME) (Definition 1.1.1). Although still relatively underexplored, this approach seems particularly promising to us. In ME, the goal is to understand how information is represented, transformed, and propagated within the NN. From this perspective, explainability is not achieved despite the use of NNs, but through them.

Definition 1.1.1: Mechanistic Explainability

ME is the approach that seeks to explain the behavior of an NN by analyzing its internal mechanisms, that is, how information is represented, transformed, and propagated within its NN.

To understand how NNs work internally, ME studies the intermediate representations learned by these networks. These representations are referred to as Latent Representations (LRs) [Bengio et al., 2014]. Analyzing their structure provide insights into how information relevant to decision-making is organized.

To develop an XRL method based on ME, we need to address several research questions, which are detailed in the following section.

1.2 Research Questions

To develop an XRL method based on ME, we need to answer the research questions developed in this section.

The first question concerns the analysis of LRs themselves. Numerous methods have been proposed to analyze the Latent Spaces (LSs) learned by NNs. These methods originate from different research communities and are rarely compared within a common framework. Yet each of them relies, often implicitly, on assumptions about how concepts are structured within the LS. These assumptions determine which structures a method is able to detect and which it will necessarily miss. Making them explicit is therefore a prerequisite for choosing a method suited to the analysis of the internal representations of RL agents. This observation leads to our first research question:

Research Question 1

What approaches have been proposed in the literature to analyze the internal workings of NNs?

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134 The second question concerns the organization of existing research in XRL.
 135 Several taxonomies have been proposed to organize existing XRL methods
 136 [Puiutta and Veith, 2020, Milani et al., 2024]. However, these taxonomies often
 137 rely on approaches initially developed for explainability in supervised settings.
 138 They therefore tend to apply XAI categories to RL without considering its
 139 specific characteristics. As a result, the organization of existing XRL methods
 140 remains ambiguous, making it difficult to structure the field. This observation
 141 leads to our second research question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL?

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143 The analysis of existing XRL approaches also reveals a third limitation.
 144 Among existing methods for explaining RL agents, few adopt a mechanistic per-
 145 spective. Understanding LR provide insights into how information is represented
 146 within the network and, in turn, help explain agent behavior. This observation
 147 leads to our third research question:

Research Question 3

How can the internal representations of NNs be exploited to identify the mechanisms underlying the decision-making process of RL agents?

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149 The analysis of existing XRL methods highlights a fourth question concerning
 150 the evaluation of explanations. Although numerous approaches have been
 151 proposed to explain the behavior of RL agents, there is no widely accepted set of
 152 objective criteria for evaluating the quality and relevance of the explanations they
 153 produce. Existing evaluations often rely on user studies or qualitative analyses,
 154 making it difficult to systematically assess the validity of an explanation approach
 155 [Doshi-Velez and Kim, 2017, Milani et al., 2024].

156 This issue is particularly important for approaches that seek to explain agent
 157 behavior by analyzing LR. Multiple hypotheses can be proposed about how
 158 information is structured within a LS. It is therefore important to define objective
 159 criteria for comparing these competing hypotheses. This leads to our fourth
 160 research question:

Research Question 4

How can the quality and relevance of explanations produced for RL agents be objectively quantified?

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162 Together, these four research questions define the structure of this manuscript.
 163 Research Question 1 addresses the organization of existing LS analysis methods
 164 and identifies the assumptions on which each of them relies. Research Question 2
 165 addresses the organization of the existing XRL literature. Research Question 3

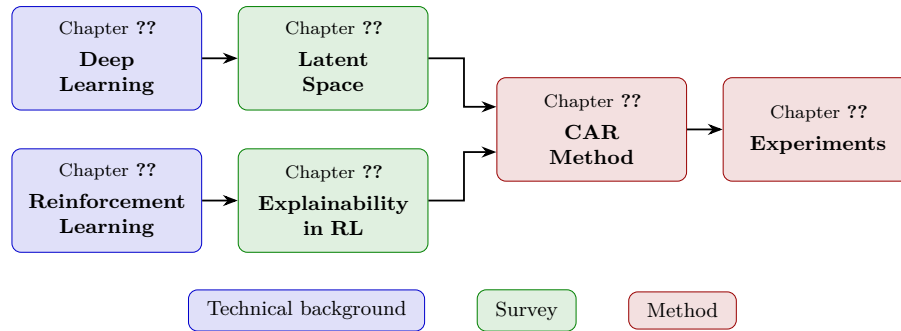


Figure 1.1: Structure of the manuscript.

166 builds on these two analyses by investigating how the internal representations of
 167 NNs can be exploited to provide mechanistic explanations of RL agent behavior.
 168 Research Question 4 addresses the complementary problem of determining how
 169 such explanations can be evaluated objectively.

170 1.3 Contributions

171 To address these four research questions, this manuscript is based on three main
 172 contributions. Two of them have been published in peer-reviewed conference,
 173 while the third is, for the moment, specific to this manuscript.

174 The first contribution addresses Research Question 1. It has not yet been
 175 published and is, for the moment, specific to this manuscript. In this contribution,
 176 we make explicit the working hypothesis that implicitly underlies much of the
 177 literature on LS analysis, which we refer to as the *Concept Hypothesis* (Hyp. ??).
 178 We then organize existing LS analysis methods into four families, namely prob-
 179 ing, interpretable directions, superposition, and clustering, according to the
 180 assumptions they make about the Concept Structure (CS).

181 The second contribution addresses Research Question 2. This contribution
 182 takes the form of a first paper [Alaarabiou et al., 2024] and an extended English
 183 version [Alaarabiou et al., 2026b], both of which were submitted for review and
 184 accepted for publication. In this contribution, we first clarify what explainability
 185 means in RL by providing a clear frame of reference. Building on these frame,
 186 we then conduct a comprehensive analysis of the state of the art in XRL.

187 The third contribution addresses Research Questions 3 and 4. In [Alaarabiou
 188 et al., 2026a], we introduce Clustering Agent Representation (CAR), an XRL
 189 method based on the principles of ME. By analyzing the LRs, CAR produces
 190 explanations in the form of clusters of states that are similar from the agent’s
 191 own perspective. We further show that this representation remains stable across
 192 training, indicating that the agent develops an objective representation of a
 193 given task.

194 1.4 Manuscript Structure

195 This manuscript is an account of the research conducted throughout this work,
 196 written for the reviewers and jury members who will evaluate it. It is nevertheless

intended to remain as accessible as possible to readers with a background in AI. To this end, the background chapters introduce and recall the concepts on which the subsequent chapters rely, allowing readers to follow the manuscript without having to consult external material. The manuscript is organized into a sequence of chapters, each addressing a specific topic related to the research questions. Each chapter falls into one of these three categories according to its objective:

Technical background chapters introduce fundamental concepts and established techniques required to understand the subsequent contributions.

Survey chapters review the existing literature around a specific research question. The organization adopted in each of these surveys are original and therefore constitute contributions of this manuscript.

Method chapters present CAR, our ME method for RL.

The chapters of the manuscript are organized as follows:

Chapter ?? Deep Learning

This chapter introduces the fundamental concepts of DL and artificial NNs.

Chapter ?? Latent Space Analysis

This chapter addresses Research Question 1b yprovides a survey of existing approaches for analyzing the LSs.

Chapter ?? Reinforcement Learning

This chapter introduces the fundamental concepts of RL.

Chapter ?? Explainability in Reinforcement Learning

This chapter addresses Research Question 2 by reviews existing approaches to explainability in RL.

Chapter ?? The CAR Method

Research Questions 3 and 4 are addressed by proposing a new method for explainability in RL based on the analysis of LRs.

Chapter ?? Experimental Evaluation of the CAR Method

This chapter presents the implementation and experimental evaluation of the proposed method.

Conclusion

The final chapter summarizes the main contributions of the manuscript, discusses the limitations of the proposed approaches, and presents possible directions for future research.

Now that the research context and objectives have been presented, we can turn to the first chapter, which introduces the fundamentals of DL.

232 Acronyms

233 **AI** Artificial Intelligence 3, 7

234 **CAR** Clustering Agent Representation 6, 7

235 **CS** Concept Structure 6

236 **DL** Deep Learning 3, 7

237 **LR** Latent Representation 4–7

238 **LS** Latent Space 4–7

239 **ME** Mechanistic Explainability 4, 6, 7

240 **NN** Neural Network 3–7

241 **RL** Reinforcement Learning 3–7

242 **XAI** eXplainable Artificial Intelligence 4, 5

243 **XRL** eXplainable Reinforcement Learning 4–6

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